CS 341: Foundations of Computer Science II
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Homework 8

1. Consider the decision problem of testing whether a DFA and a regular expression are equivalent. Express this problem as a language and show that it is decidable.

2. Consider the decision problem of testing whether a CFG generates the empty string. Express this problem as a language and show that it is decidable.

3. Let $\Sigma = \{0, 1\}$, and consider the decision problem of testing whether a regular expression with alphabet Σ generates at least one string w that has 111 as a substring. Express this problem as a language and show that it is decidable.

4. Consider the emptiness problem for Turing machines:

   $$E_{\text{TM}} = \{ \langle M \rangle \mid M \text{ is a Turing machine with } L(M) = \emptyset \}.$$

   Show that E_{TM} is co-Turing-recognizable. (A language L is co-Turing-recognizable if its complement $\overline{L}$ is Turing-recognizable.) Note that the complement of E_{TM} is

   $$\overline{E_{\text{TM}}} = \{ \langle M \rangle \mid M \text{ is a Turing machine with } L(M) \neq \emptyset \}.$$

   (Actually, $\overline{E_{\text{TM}}}$ also contains all $\langle M \rangle$ such that $\langle M \rangle$ is not a valid Turing-machine encoding, but we will ignore this technicality.)